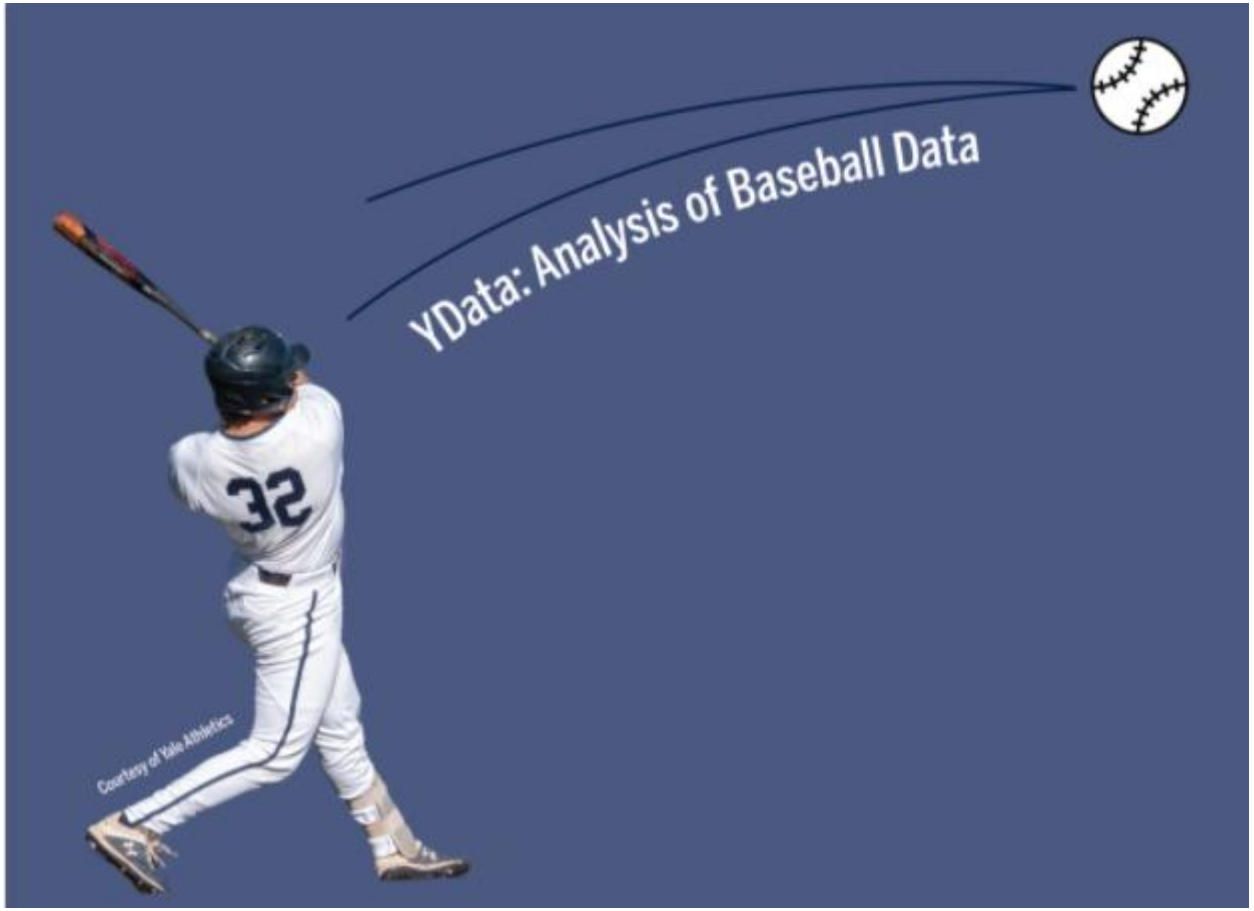
S&DS 1730



Class 7: Probability and simulations using games part II

Overview

Discussion of chapter 5 of Astroball

Discuss All-Star baseball and simulations in Python

Play Strat-O-Matic

Quick review or probability rules and Tree Diagrams

Announcement: Midterm exam

Will be in person next class

- Monday March 2nd

The exam will be in a Jupyter notebook and you will fill in the notebook and turn along with a pdf on Gradescope

- **Be sure to bring your computer to the exam!**

You can look at previous labs, and create one additional notebook that has code in it, but the exam is otherwise closed book/Internet

Announcement: Lab 6

Lab 6 has been released. It is due on **Sunday March 22nd**

- **i.e., not due until the Sunday at the end of spring break**
  - Although I would encourage you to complete it sooner

Lab 5: Questions?

How did it go?

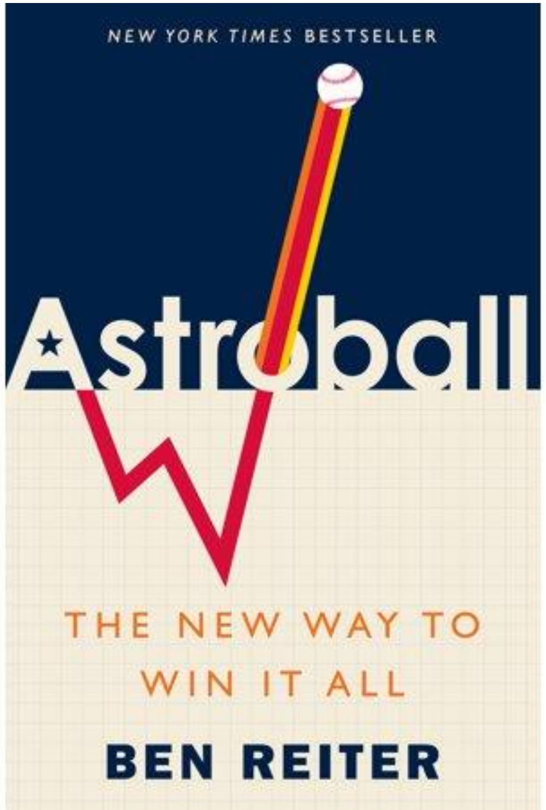
You will get a little more practice with OOP on lab 5

- You will add a method display_game() to your Baseball_Game object
- If you weren't able to compete the Baseball_Game object in lab 5, email me and I will send you the code

Discussion on chapter 5 of Astroball

Let's discuss in groups of 3-4 people:

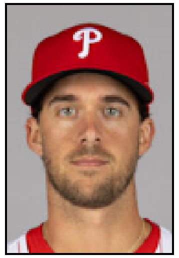
1. Chapter 5 of Astroball
    - E.g., Discuss your quote and reaction to chapter 5



Prospects Astros were considering drafting 1-1

Astros considering 6 prospects

- Less seriously: Aaron Nola, Nick Gordon



Aaron Nola

Position: Pitcher

Bats: Right • **Throws:** Right

6-2, 200lb (188cm, 90kg)

Team: Philadelphia Phillies (majors)

More bio, uniform, draft, salary info ▼

SUMMARY	WAR	W	L	ERA	G	GS	SV	IP	SO	WHIP
2025	-0.3	5	10	6.01	17	17	0	94.1	97	1.346
Career	34.9	109	89	3.83	285	285	0	1715.2	1876	1.149

Nick Gordon

Positions: Outfielder, Second Baseman and Shortstop

Bats: Left • **Throws:** Right

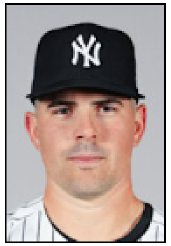
6-0, 160lb (183cm, 72kg)

Born: October 24, 1995 (Age: 30-120d) in Avon Park, FL

More bio, uniform, draft, salary info ▼

SUMMARY	WAR	AB	H	HR	BA	R	RBI	SB	OBP	SLG	OPS	OPS+
Career	-0.6	956	233	23	.244	106	112	21	.283	.386	.669	86

Prospects Astros were considering drafting 1-1

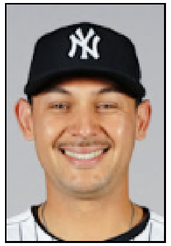


Carlos Rodón

Position: Pitcher
Bats: Left • **Throws:** Left
6-2, 255lb (188cm, 115kg)
Team: New York Yankees (majors)

More bio, uniform, draft, salary info ▼

SUMMARY	WAR	W	L	ERA	G	GS	SV	IP	SO	WHIP
2025	4.6	18	9	3.09	33	33	0	195.1	203	1.049
Career	22.3	93	72	3.73	231	226	0	1282.0	1409	1.218

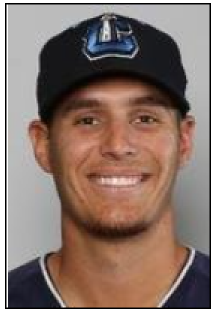


Alex Jackson

Position: Catcher
Bats: Right • **Throws:** Right
6-1, 238lb (185cm, 107kg)
Team: Minnesota Twins (majors)

More bio, uniform, draft, salary info ▼

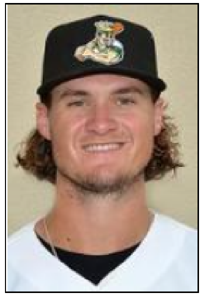
SUMMARY	WAR	AB	H	HR	BA	R	RBI	SB	OBP	SLG	OPS	OPS+
2025	0.6	91	20	5	.220	17	8	0	.290	.473	.763	111
Career	-0.7	393	60	11	.153	47	32	1	.239	.288	.527	46



Brady Aiken

Position: Starting Pitcher
Bats: Left • **Throws:** Left
6-4, 205lb (193cm, 92kg)
Team: Cleveland Indians (minors)

More bio, uniform, draft, sal



Tyler Kolek

Position: Pitcher
Bats: Right • **Throws:** Right
6-5, 260lb (196cm, 117kg)
Born: December 15, 1995 (Age: 25-090d) in Shepherd, TX
Draft: Drafted by the Miami Marlins in the 1st round (2nd) of the 2014 from Shepherd HS (Shepherd, TX).
High School: Shepherd HS (Shepherd, TX)
Full Name: Tyler Frank Kolek
Twitter: @tylerkolek

DEADSPIN

HOME LATEST NBA MLB NFL NHL SOCCER GLOBAL PANDEMIC THE MOURNING AFTER

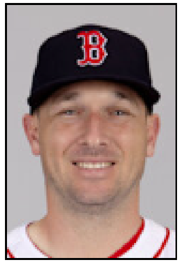
MLB

The Astros Are Trying To Dick Draft Picks Out Of Their Money

Tom Ley
7/15/14 11:58AM

2

[2014 daft results]



Alex Bregman

Positions: Third Baseman and Shortstop
Bats: Right • **Throws:** Right
5-11, 190lb (180cm, 86kg)
Team: Chicago Cubs (majors)

More bio, uniform, draft, salary info ▼

SUMMARY	WAR	AB	H	HR	BA	R	RBI	SB	OBP	SLG	OPS	OPS+
2025	3.5	433	118	18	.273	64	62	1	.360	.462	.821	128
Career	43.1	4590	1250	209	.272	758	725	43	.365	.481	.846	132

Probability

Probability models assigns a number between 0 and 1 to the outcome of an event occurring

P(event) = 0 if there is no chance of an event occurring

P(event) = 1 if the event will definitely occur

P(event) ϵ [0, 1] if there is some possibility that an event will occur

Probability

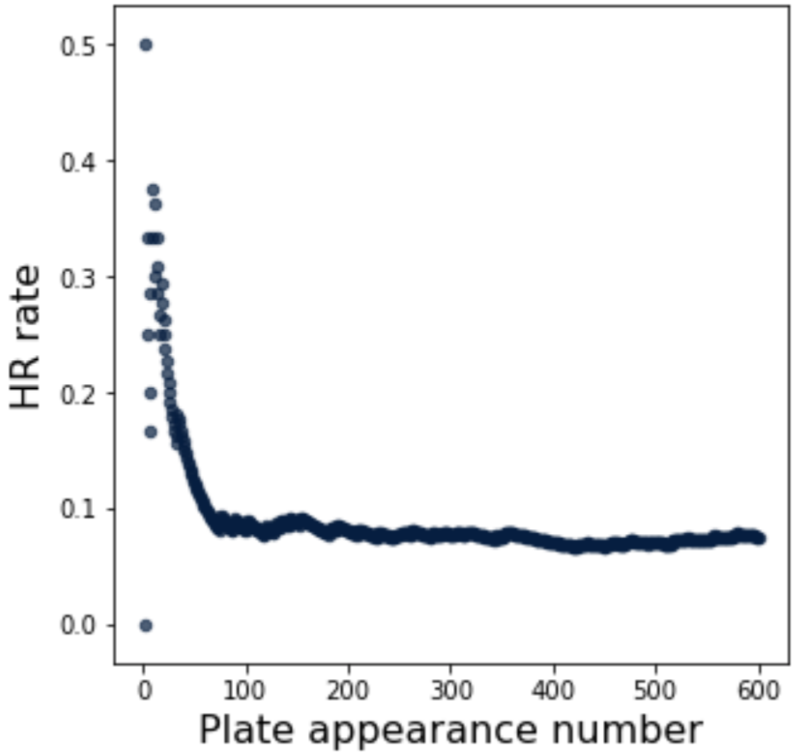
One way to interpret probability is in terms of the **relative frequency** of an event

If we repeat an experiment N times, we get:

$$P(HR) \approx \frac{\text{Number of HRs}}{\text{Number of PAs}}$$

If we repeated this infinitely many times, we would get the true probability of the event

- i.e., $N \rightarrow \infty$



Probability through estimating the relative frequency of events

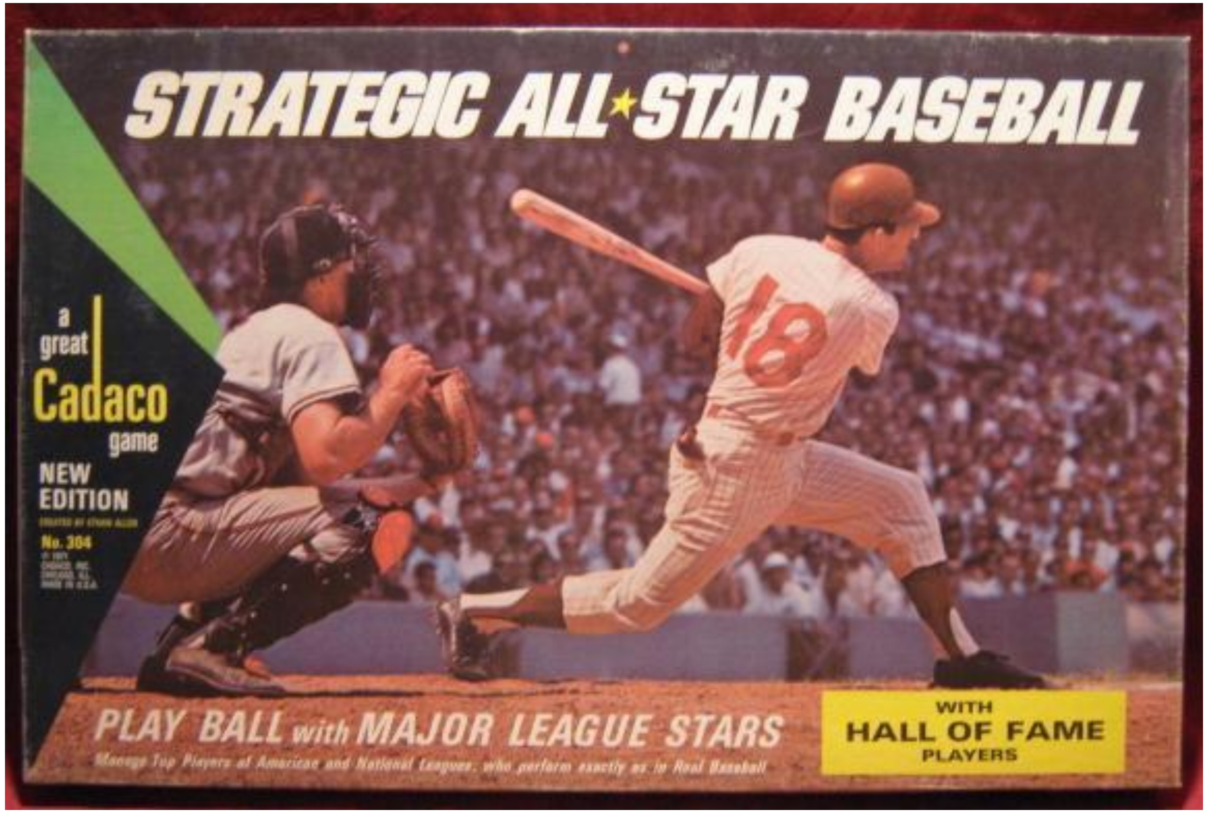
Last class we discussed calculating probabilities using the additive and multiplicative probability rules

- We will review and extend this later in the class

However sometimes it is not possible to mathematically calculate the probability distribution

In such cases, we can estimate the probability of events by counting their relative frequencies

All-Star Baseball



Game that was popular in the 1960's and 70's

All-Star Baseball

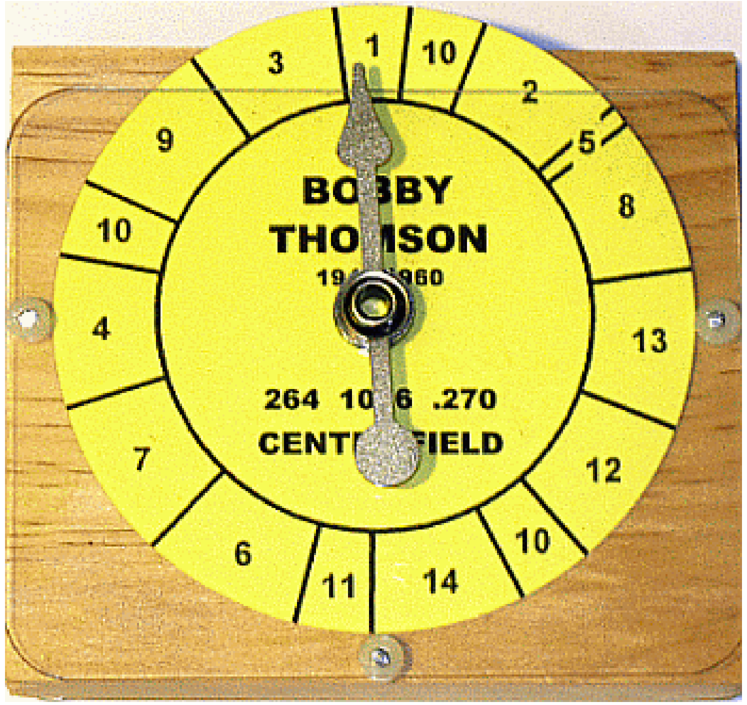
Each player represented by a spinner:

1 = home run

2, 13 = single

11 = double

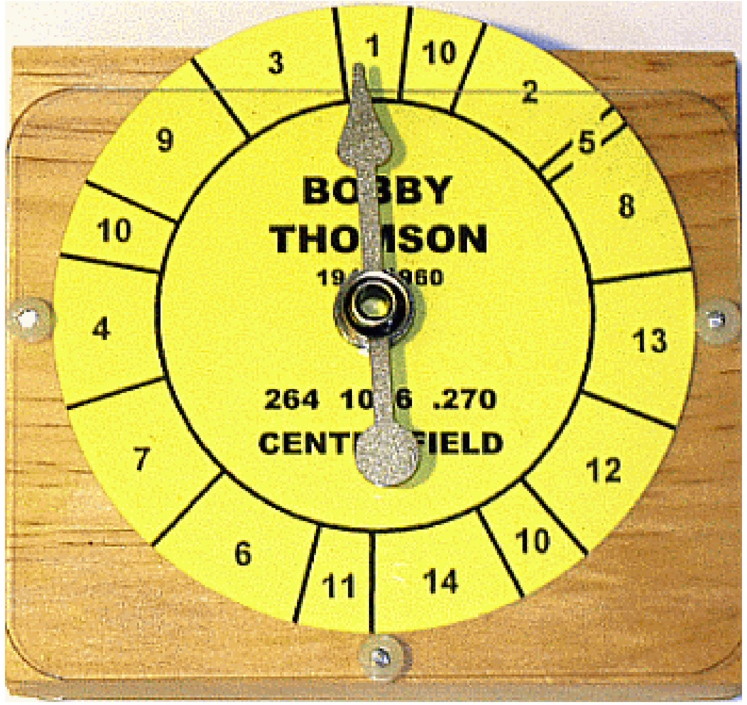
etc.



All-Star Baseball

For all-star baseball is it not possible to analytically calculate the probability of different events

- Instead, we would just have to repeat the event many times can calculate the relative frequencies



All-Star Baseball

Unfortunately, due to snow we are not going to play All-Star Baseball

But you will simulate it in Python on lab 6!

Simulating All-Star Baseball in Python

Steps to simulate a game of All-Star Baseball in Python?

1. get_spinner_probabilities(retro_id)
2. get_lineups(game_id)
3. get_all_spinners(away_lineup, home_lineup)
4. simulate_game(game_id)

Simulating All-Star Baseball in Python

1. spinner = get_spinner_probabilities('ohtas001')

Returns a Series

Index is event name

```
out        0.602469
single     0.111111
double     0.034568
triple     0.012346
hr         0.077778
walk       0.158025
roe        0.003704
dtype: float64
```

Values are proportion of times the event occurred

curr_play = np.random.choice(spinner.index, p = spinner)

Simulating All-Star Baseball in Python

2. get_lineups(game_id)

- For a given game_id, find the retro_ids for first 9 home and away batters

- Returns a tuple (away_lineup, home_lineup)
  - away_lineup and home_lineup are DataFrames

- Uses just the first 9 players who batted for each team for the given game_id

BAT_ID
calhk001
troum001
bourj002
simma001
pujoa001
lastt001
lucrj001
goodb001
cozaz001

Simulating All-Star Baseball in Python

3. get_all_spinners(away_lineup, home_lineup)

- Returns a two lists as a tuple: (away_spinners, home_spinners)
- Each list, contains the spinner Series for the 9 players in the line up

```
home_spinners = []
for i in range(0,9):
    # stuff
    home_spinners.append( …)
```

Simulating All-Star Baseball in Python

4. simulate_game(game_id)

- Uses a Baseball_Game() object to keep track of the game and loops until the end of the game
- Keep track of the current batter (for both the home and away team)
- Gets the spinner DataFrame for the current batter
- Generates a random play based on the spinner probabilities
- Has a series of *if statements* to tell how to update the Baseball_Game state

Simulating All-Star Baseball in Python

Questions?

Strat-o-matic

Strat-O-Matic™
50TH
1961 – 2011
ANNIVERSARY
THE ORIGINAL FANTASY SPORTS GAMES!

Much more complex board games

- Takes into account Hitters and Pitchers
- Advanced version accounts for additional factors (e.g., ball parks etc.)

Strat-o-matic

NELSON CRUZ rightfield-2 leftfield-3 stealing-A running 1-12

TEXAS

1

2-flyball (cf)B
3-groundball (3b)A
4-groundball (3b)B
5-strikeout
6-strikeout
7-WALK
8-groundball (ss)A
9-strikeout
10-groundball (p)B
11-groundball (ss)A++
12-lineout (2b) into as many outs as possible

2

2-flyball (cf)A
3-groundball (ss)A++
4-groundball (ss)A
5-strikeout
6-strikeout
7-flyball (lf)B
8-strikeout
9-strikeout
10-flyball (cf)B
11-groundball (ss)A++ plus injury
12-WALK

3

2-SINGLE* 1-2 lineout (3b) 3-20
3-WALK
4-groundball (3b)B
5-SINGLE
6-HOMERUN
7-HOMERUN 1-10 DOUBLE 11-20
8-groundball (3b)A
9-DOUBLE** 1 SINGLE** 2-20
10-HOMERUN
11-SINGLE
12-flyball (rf)B

2009 BATTING RECORD

AVG	AB	2B	3B	HR	RBI
.260	462	21	1	33	76

BB	SO	SB	CS	SLG%	ON BASE %
49	118	20	4	.524	.332

BRANDON McCARTHY pitcher-3 batting # 1 starter

PITCHING CARD **TEXAS**

4

2-groundball (p)B
3-FLYBALL (lf)X
4-flyball (cf)B
5-groundball (2b)C
6-SINGLE* 1-10 lineout (3b) 11-20
7-flyball (rf)B
8-DOUBLE** 1-12 SINGLE** 13-20
9-flyball (lf)C
10-popout (1b)
11-FLYBALL (rf)X
12-flyball (lf)B

5

2-lineout (ss)
3-TRIPLE 1 DOUBLE 2-20
4-CATCHER'S CARD X
5-HOMERUN 1-19 DOUBLE 20
6-strikeout
7-GROUND-BALL(2b)X
8-flyball (cf)B
9-strikeout
10-GROUND-BALL(ss)X
11-strikeout
12-groundball (1b)C

6

2-groundball (1b)C
3-GROUND-BALL(p)X
4-GROUND-BALL(3b)X
5-strikeout
6-WALK
7-SINGLE
8-WALK
9-GROUND-BALL(ss)X
10-FLYBALL (cf)X
11-GROUND-BALL(1b)X
12-groundball (p)B

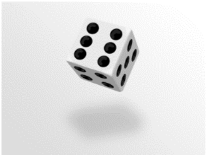
2009 PITCHING RECORD

W	L	ERA	STARTS	SAVES
7	4	4.62	17	0

IP	HITS ALLOWED	BB	SO	HOMERUNS ALLOWED
97	96	36	65	13

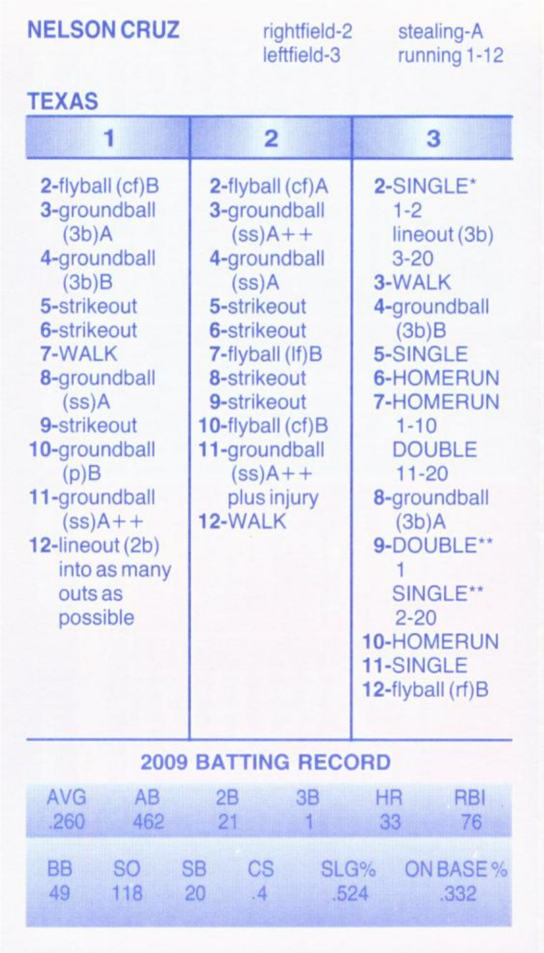
Each player is represented by a card

A single white die is rolled to determine whether to use hitter or pitcher's card:



- 1-3 -> hitters card
- 4-6 -> pitcher's card

Strat-o-matic



NELSON CRUZ
rightfield-2
leftfield-3
stealing-A
running 1-12

TEXAS

1	2	3
2-flyball (cf)B 3-groundball (3b)A 4-groundball (3b)B 5-strikeout 6-strikeout 7-WALK 8-groundball (ss)A 9-strikeout 10-groundball (p)B 11-groundball (ss)A++ 12-lineout (2b) into as many outs as possible	2-flyball (cf)A 3-groundball (ss)A++ 4-groundball (ss)A 5-strikeout 6-strikeout 7-flyball (lf)B 8-strikeout 9-strikeout 10-flyball (cf)B 11-groundball (ss)A++ plus injury 12-WALK	2-SINGLE* 1-2 lineout (3b) 3-20 3-WALK 4-groundball (3b)B 5-SINGLE 6-HOMERUN 7-HOMERUN 1-10 DOUBLE 11-20 8-groundball (3b)A 9-DOUBLE** 1 SINGLE** 2-20 10-HOMERUN 11-SINGLE 12-flyball (rf)B

2009 BATTING RECORD

AVG	AB	2B	3B	HR	RBI
.260	462	21	1	33	76

BB	SO	SB	CS	SLG%	ON BASE %
49	118	20	4	.524	.332

Then, two red dice are rolled and their sum determines which play in the card should be used

For some plays additionally a 20 sided die is rolled to determine the final outcome

- and other tables/rules often need to be consulted

Strat-o-matic

Let’s play!

1. Divide into two teams: Rays and Cardinals

2. One person in the class needs to be selected for each player:

Enter your batting order here: http://bit.ly/strat-o-matic

We will use the dice simulators on Canvas

Someone keep score?

Strat-o-matic

We are going to keep it simple so we will ignore a few rules…

- No base stealing, ignore ++
- For batter's card, we can ignore which position the ball was hit to
  - E.g., pay attention to flyball A, ignore (rf)
- For pitcher's card, when there is an X we will use the position (card) the ball was hit to and refer to a table

Batting will roll all the dice…

- Except for fielding plays

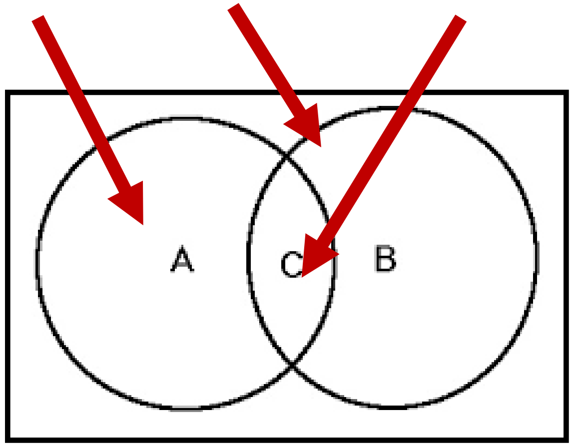
Play ball!

Probability rules

Probability rules - Additive rule

If there are two events A, and B, then the probability of A *or* B happening is:

P(A or B) = P(A) + P(B) - P(A, B)



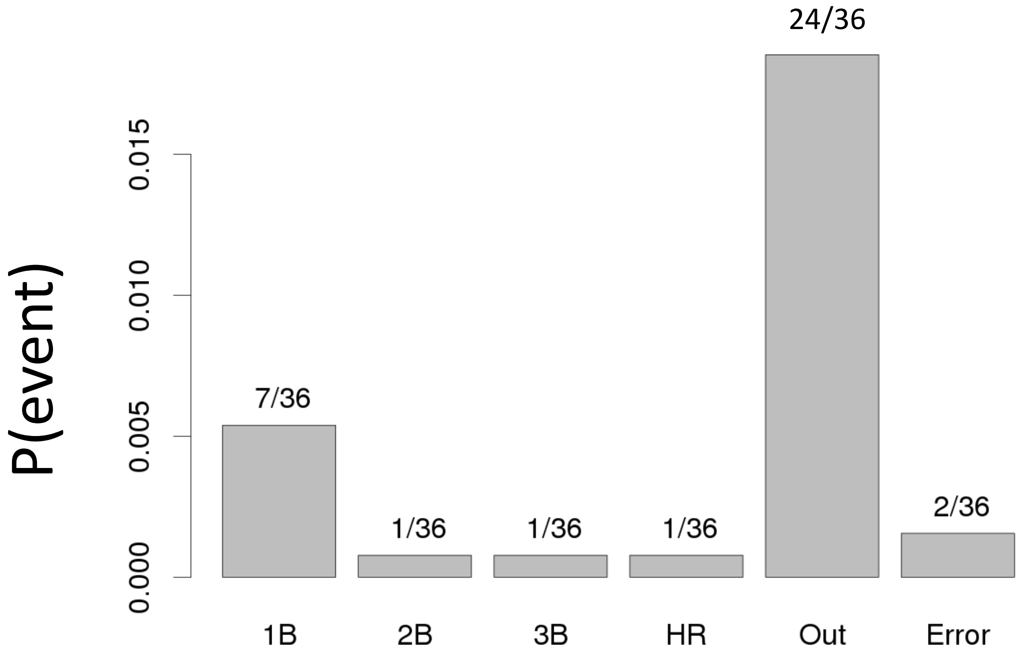
Events are called **mutually exclusive** if events A and B can not both occur; i.e., P(A, B) = 0

Q: What would mutually exclusive events look like in the Venn diagram?

A: The circles would not overlap

Probability rules - Additive rule

Q: If the ball is in play, what is the probability of getting a hit in big league baseball?



P(hit) =

= P(1B or 2B or 3B or HR)

= P(1B) + P(2B) + P(3B) + P(HR)

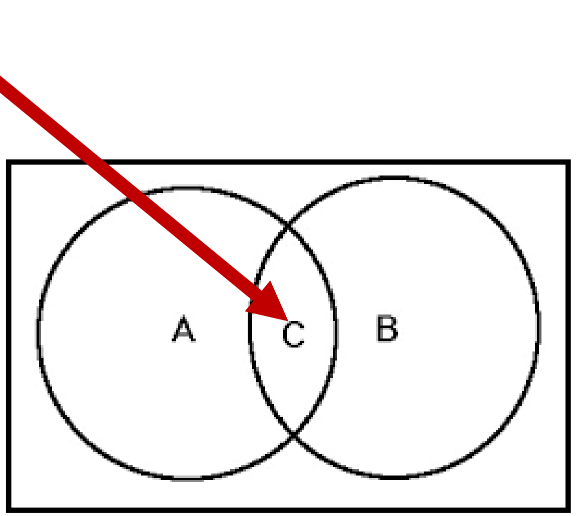
= 7/36 + 1/36 + 1/36 + 1/36

= 10/36

Multiplicative Rule

The probability of two events A ***and*** B occurring is:

$$P(A, B) \quad = \quad P(A|B) \times P(B)$$



Probability of B happening

×

Probability of A happening given B happened

Special case: two events are **independent** if:

$P(A, B) = P(A) \times P(B)$

i.e., if the occurrence of B does not effect the probability of A happening

Big League Baseball

Recall that the first pitch for Big League Baseball is modeled using…

A single die is rolled:

- If a 2 or 3 occurs, a ball is pitched
- If a 4 or 5 occurs, a strike is pitched
- If a 1 or 6 occurs, <u>a ball is hit in fair play</u>

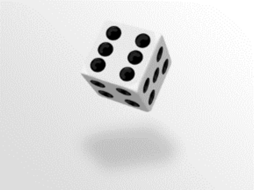
What is the probability one would strike out on three straight pitches?

- Answer: $(1/3)^3 = 1/27$

Strat-o-matic rules and tree diagrams

Each player is represented by a card:

1. A white single die is rolled to determine whether to use hitter or pitcher's card:
    - 1-3 -> hitters card
    - 4-6 -> pitcher's card



2. Then, two dice are rolled and their sum determines which play in the card should be used

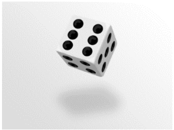
3. For some plays additionally a 20 sided die is rolled to determine the final outcome

Strat-o-matic: analysis

Let’s calculate the probability of different events…

Calculating the probability of getting a particular column in a pitcher or hitter’s card is pretty simple

- Answer?



Calculating the sum of the two dice is a little more involved…

- What is the sample space here?
  - i.e., what are the possible outcomes here?
- Can you calculate the probability distribution?

Strat-o-matic: analysis

We can fill in the table below with the sum of the two dice and then calculate the probability of rolling a 2 to a 12

2nd Die

1st Die

	1	**2**	**3**	**4**	**5**	**6**
1						
2						
3						
4			7			
5						
6						

Strat-o-matic: analysis

We can fill in the table below with the sum of the two dice and then calculate the probability of rolling a 2 to a 12

2nd Die

1st Die

	1	**2**	**3**	**4**	**5**	**6**
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Strat-o-matic: analysis

2	3	4	5	6	7	8	9	10	11	12
1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

2nd Die

1^{st} Die

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Strat-o-matic: analysis

2	3	4	5	6	7	8	9	10	11	12
1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

What is the probability of rolling a 1 on the white die **and then** getting a sum of 7 on the two red dice?

What is the probability of rolling a:

- 2 on the white die…. **and then** getting…
- Sum of 8 on the two red dice… **and then…**
- A number for 1-8 on the 20 sided die?

Strat-o-matic: analysis

2	3	4	5	6	7	8	9	10	11	12
1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

What is the probability of rolling a 1 on the white die **or** a 6 on the white die?

What is the probability of rolling a:

- 5 on the white die… **and then** getting …
- a sum of 8 **or** a sum of 10 on the two red dice?

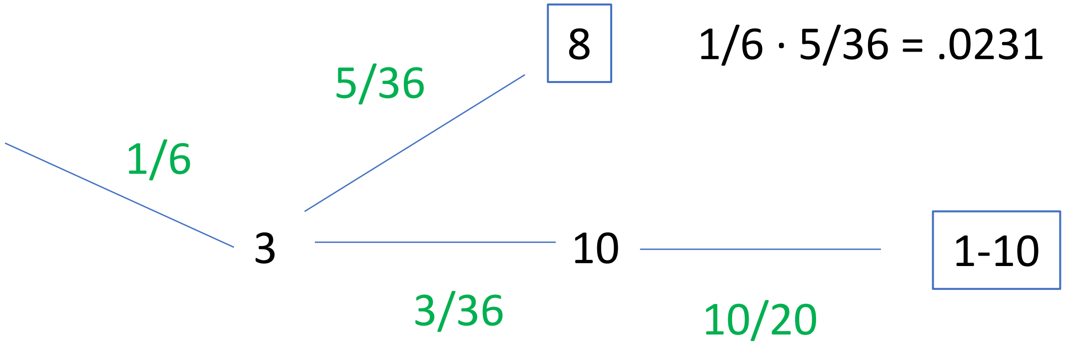
Strat-o-matic: analysis

2	3	4	5	6	7	8	9	10	11	12
1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

What is the probability of rolling a 3 on the white die

- and then getting a sum of 8 on the two red dice
- or a sum of 10 on the two red dice
    - and then a 1-10 on the 20 sided die?

Tree diagram!

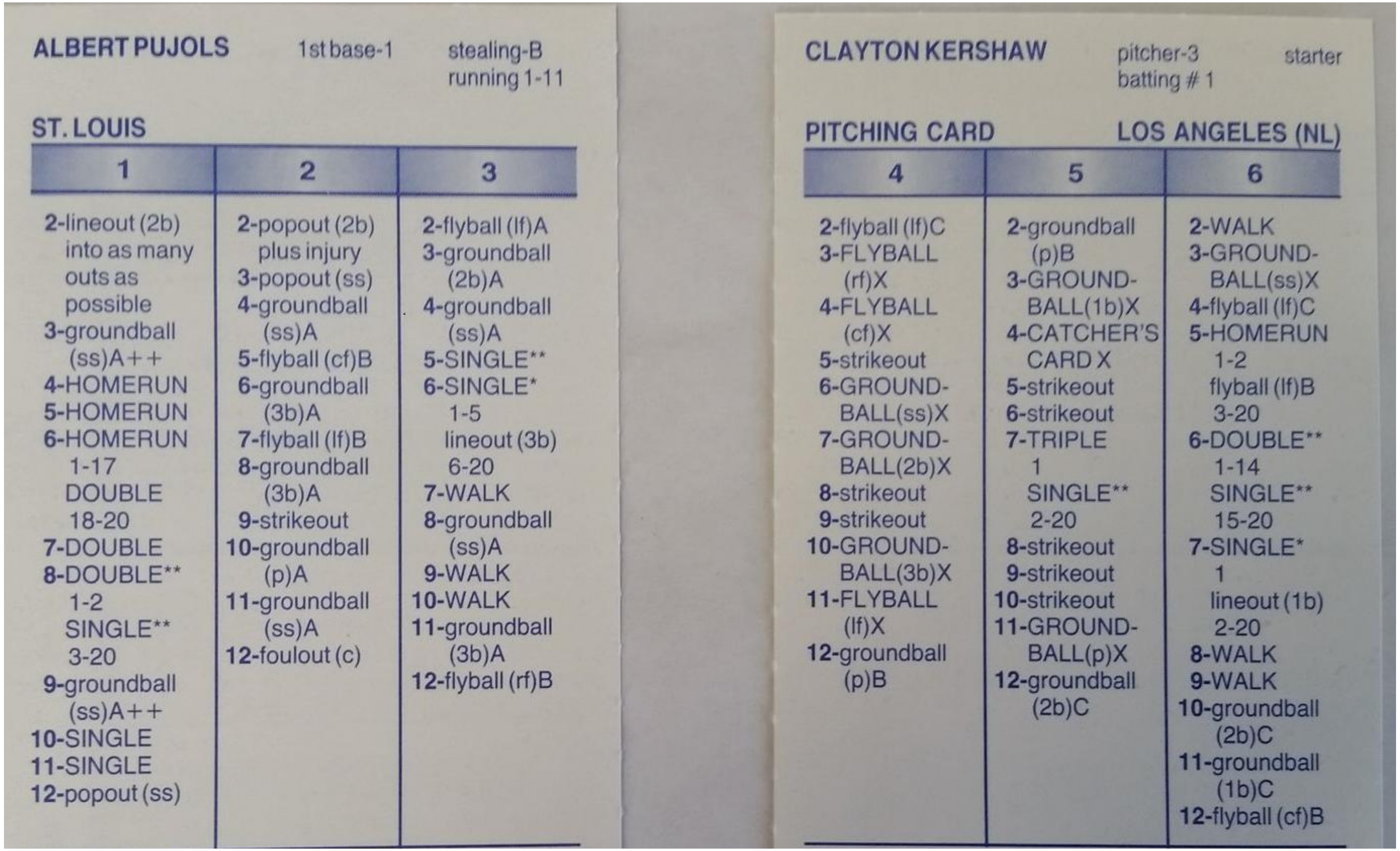


$1/6 \cdot 5/36 = .0231$

$1/6 \cdot 3/36 \cdot ½ = .0069$

.0231 + .0069 = .0300

Strat-o-matic: Pujols vs. Kershaw



ALBERT PUJOLS 1st base-1 stealing-B running 1-11

ST. LOUIS

1	2	3
2-lineout (2b) into as many outs as possible 3-groundball (ss)A++ 4-HOMERUN 5-HOMERUN 6-HOMERUN 1-17 DOUBLE 18-20 7-DOUBLE 8-DOUBLE** 1-2 SINGLE** 3-20 9-groundball (ss)A++ 10-SINGLE 11-SINGLE 12-popout (ss)	2-popout (2b) plus injury 3-popout (ss) 4-groundball (ss)A 5-flyball (cf)B 6-groundball (3b)A 7-flyball (lf)B 8-groundball (3b)A 9-strikeout 10-groundball (p)A 11-groundball (ss)A 12-foulout (c)	2-flyball (lf)A 3-groundball (2b)A 4-groundball (ss)A 5-SINGLE** 6-SINGLE* 1-5 lineout (3b) 6-20 7-WALK 8-groundball (ss)A 9-WALK 10-WALK 11-groundball (3b)A 12-flyball (rf)B

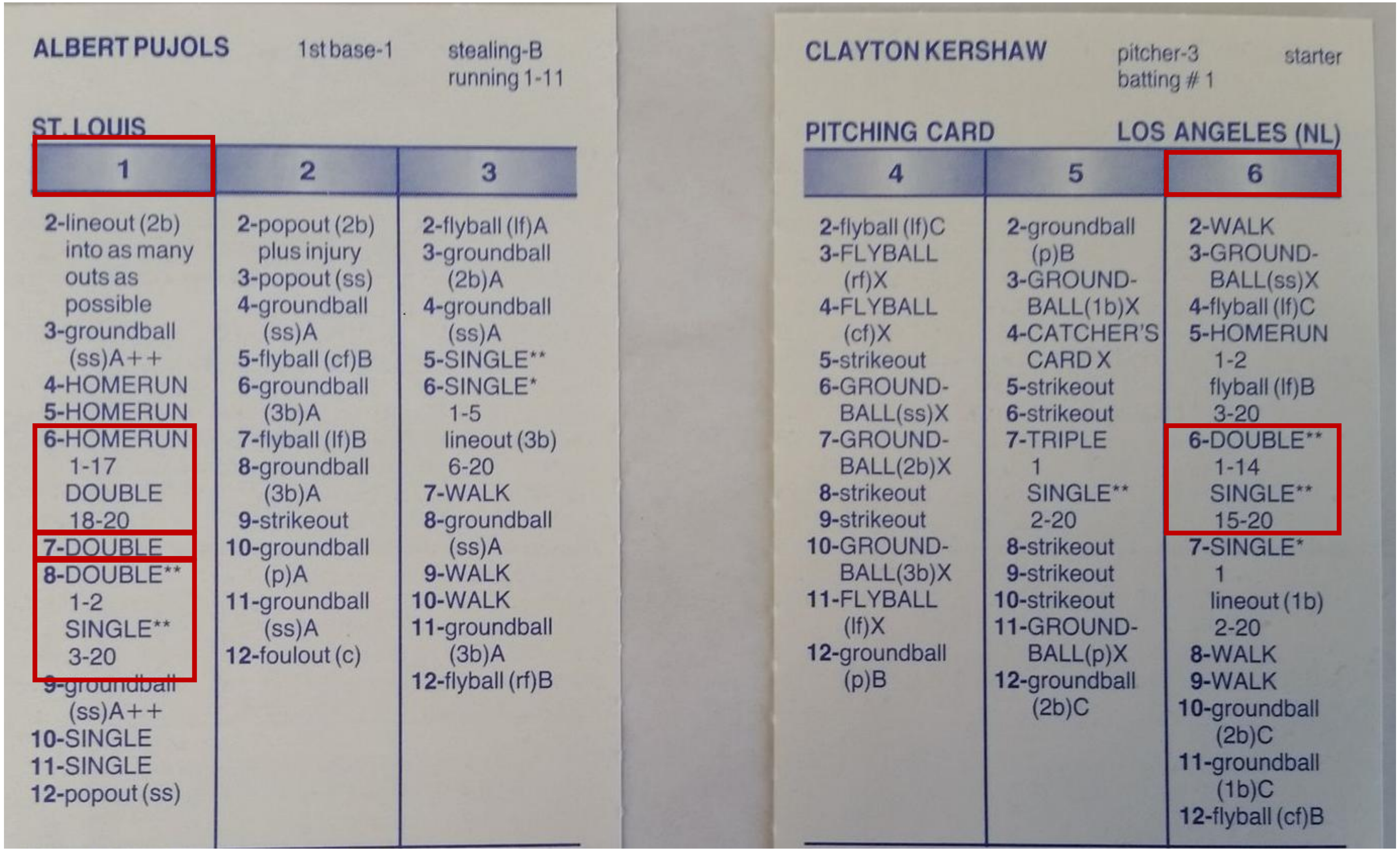
CLAYTON KERSHAW pitcher-3 batting # 1 starter

PITCHING CARD **LOS ANGELES (NL)**

4	5	6
2-flyball (lf)C 3-FLYBALL (rf)X 4-FLYBALL (cf)X 5-strikeout 6-GROUND-BALL(ss)X 7-GROUND-BALL(2b)X 8-strikeout 9-strikeout 10-GROUND-BALL(3b)X 11-FLYBALL (lf)X 12-groundball (p)B	2-groundball (p)B 3-GROUND-BALL(1b)X 4-CATCHER'S CARD X 5-strikeout 6-strikeout 7-TRIPLE 1 SINGLE** 2-20 8-strikeout 9-strikeout 10-strikeout 11-GROUND-BALL(p)X 12-groundball (2b)C	2-WALK 3-GROUND-BALL(ss)X 4-flyball (lf)C 5-HOMERUN 1-2 flyball (lf)B 3-20 6-DOUBLE** 1-14 SINGLE** 15-20 7-SINGLE* 1 lineout (1b) 2-20 8-WALK 9-WALK 10-groundball (2b)C 11-groundball (1b)C 12-flyball (cf)B

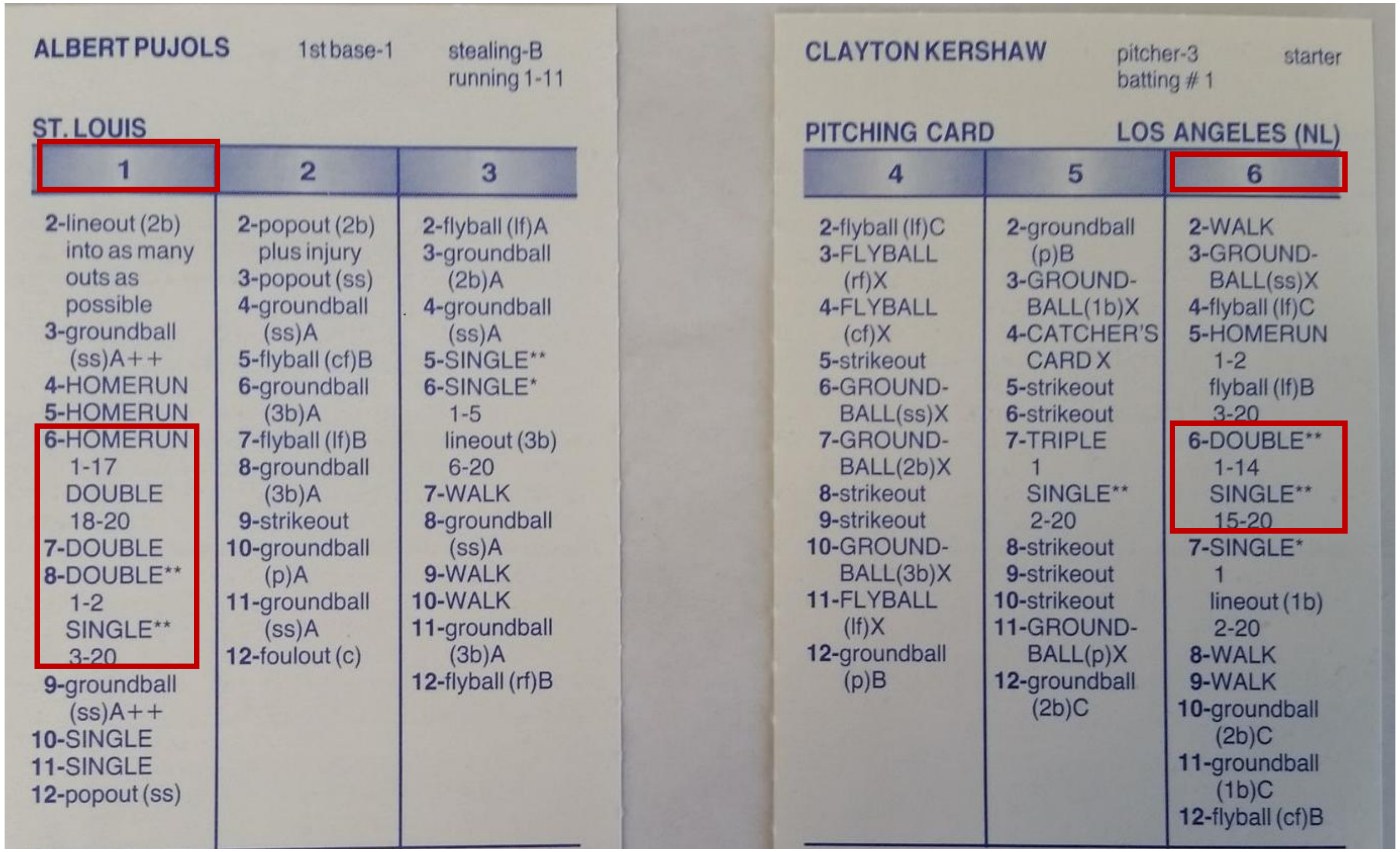
What is the probability of a hitting a double?

Strat-o-matic: Pujols vs. Kershaw



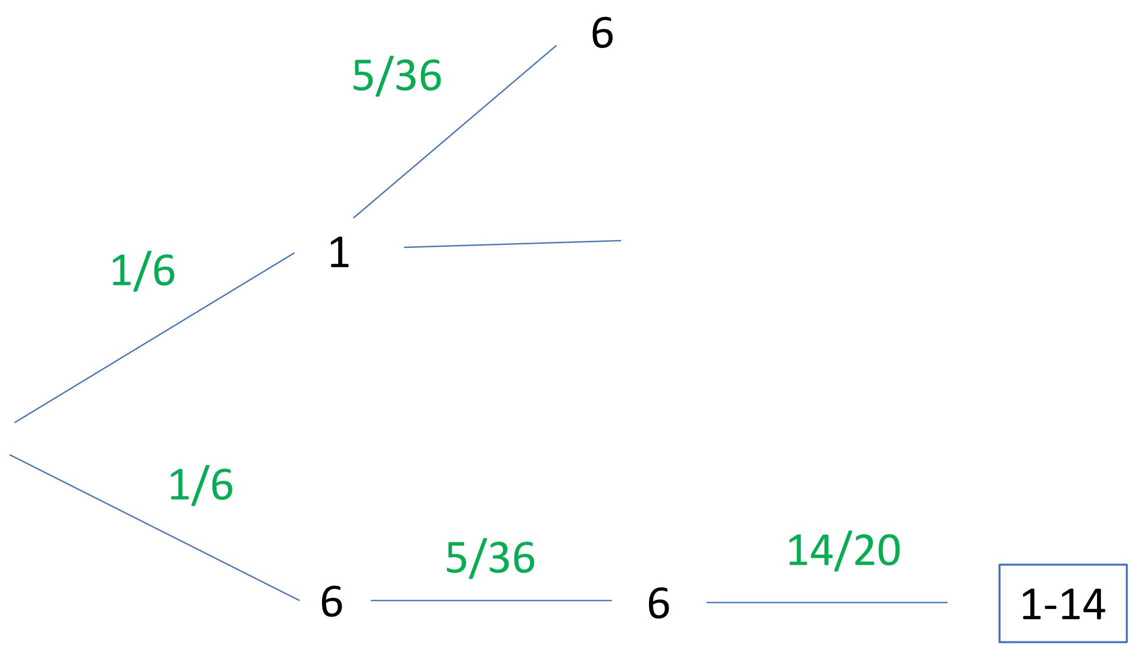
What is the probability of a hitting a double?

Strat-o-matic: Pujols vs. Kershaw

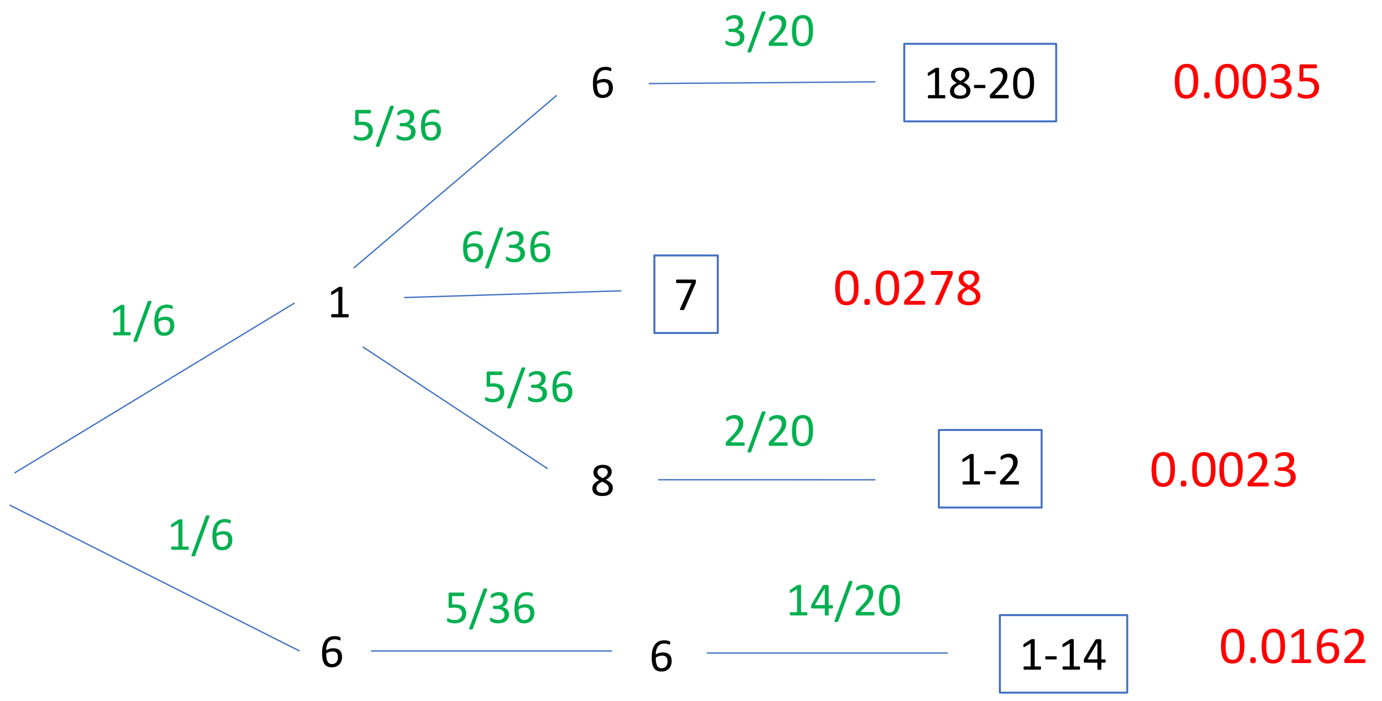


2	3	4	5	6	7	8	9	10	11	12
1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

Tree diagram

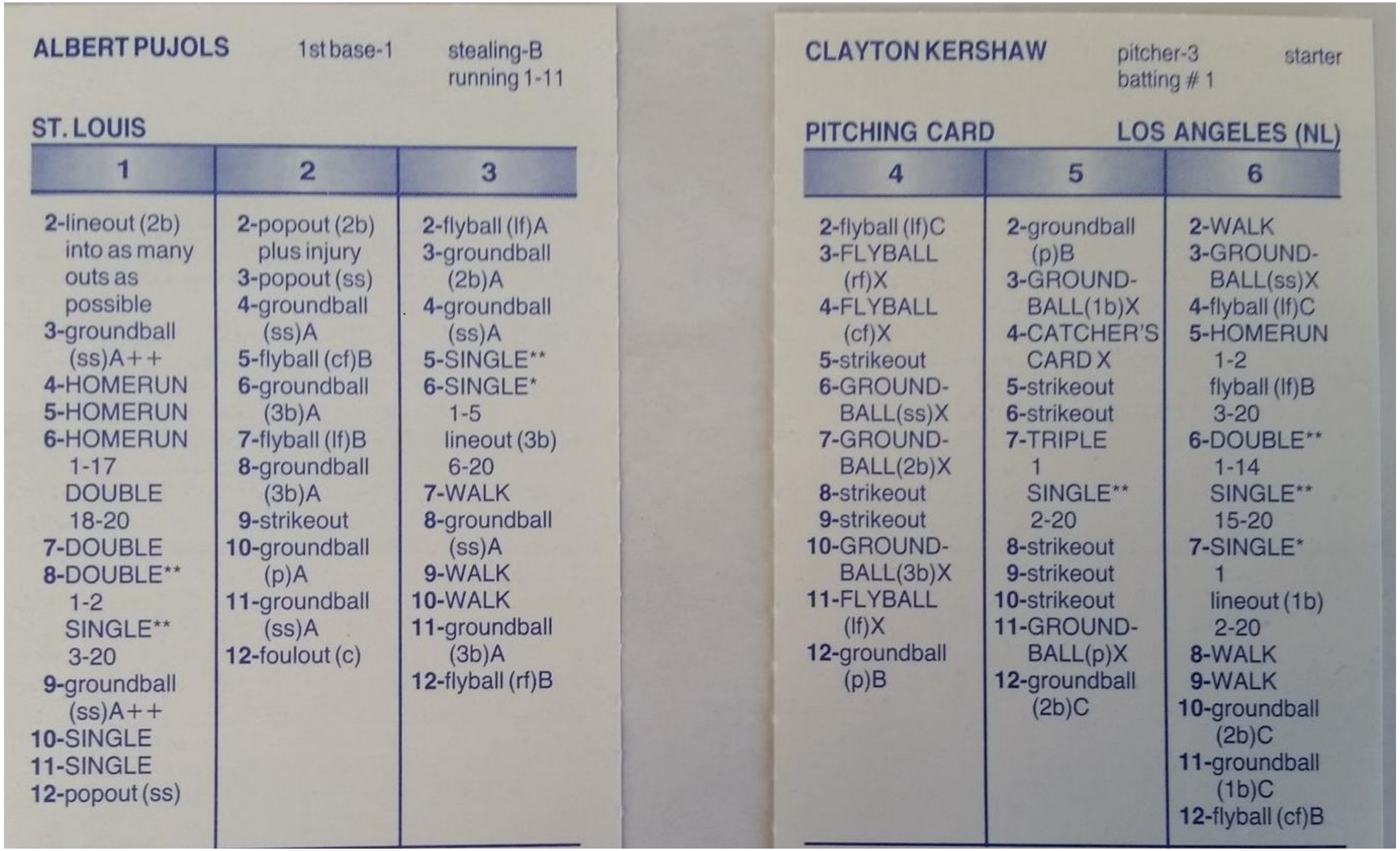


Tree diagram



.0035 + .0278 + .0023 + .0162 = .0498

Strat-o-matic: Pujols vs. Kershaw



ALBERT PUJOLS 1st base-1 stealing-B running 1-11

ST. LOUIS

1	2	3
2-lineout (2b) into as many outs as possible 3-groundball (ss)A++ 4-HOMERUN 5-HOMERUN 6-HOMERUN 1-17 DOUBLE 18-20 7-DOUBLE 8-DOUBLE** 1-2 SINGLE** 3-20 9-groundball (ss)A++ 10-SINGLE 11-SINGLE 12-popout (ss)	2-popout (2b) plus injury 3-popout (ss) 4-groundball (ss)A 5-flyball (cf)B 6-groundball (3b)A 7-flyball (lf)B 8-groundball (3b)A 9-strikeout 10-groundball (p)A 11-groundball (ss)A 12-foulout (c)	2-flyball (lf)A 3-groundball (2b)A 4-groundball (ss)A 5-SINGLE** 6-SINGLE* 1-5 lineout (3b) 6-20 7-WALK 8-groundball (ss)A 9-WALK 10-WALK 11-groundball (3b)A 12-flyball (rf)B

CLAYTON KERSHAW pitcher-3 batting # 1 starter

PITCHING CARD **LOS ANGELES (NL)**

4	5	6
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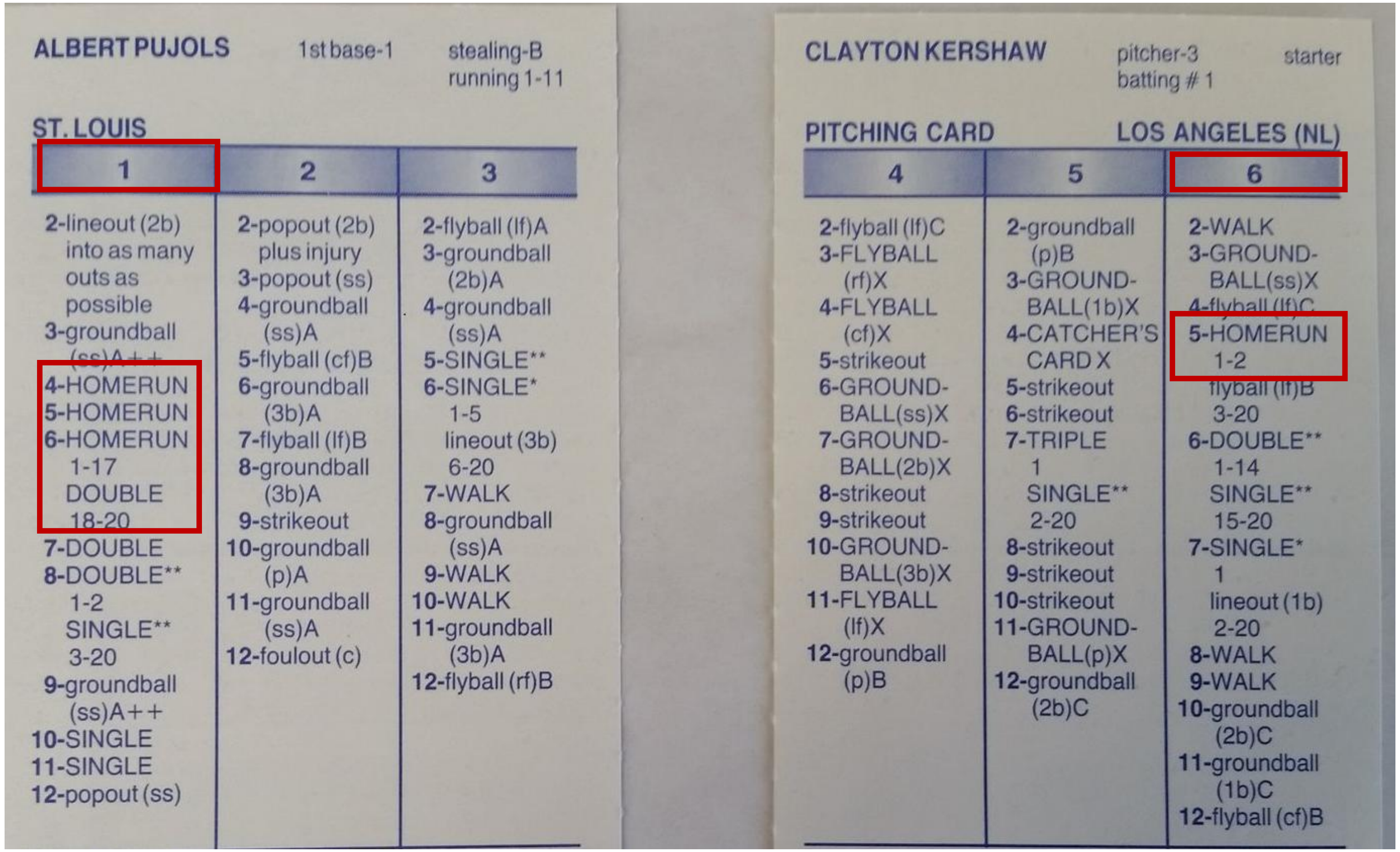
What is the probability of a hitting a home run?

Strat-o-matic: Pujols vs. Kershaw



What is the probability of a hitting a home run?   Tree diagram!

2	3	4	5	6	7	8	9	10	11	12
1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

Using probability notation with tree diagrams

$P(X_1 = 1, X_2 = 4) =$

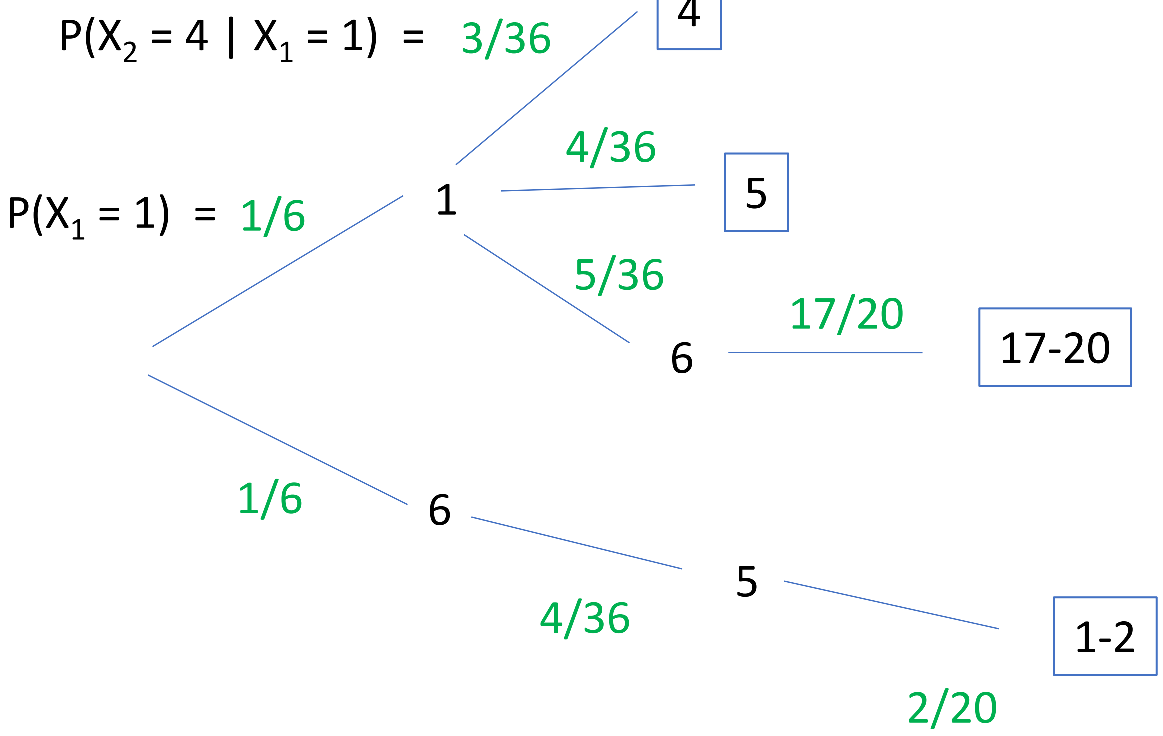
$P(X_2 = 4 \mid X_1 = 1) \cdot P(X_1 = 1) =$

$1/6 \cdot 3/36 = 0.0139$

X_1

X_2

X_3



What is the sample space for X_1 ?

Using probability notation with tree diagrams

$P(X_1 = 1, X_2 = 4) =$

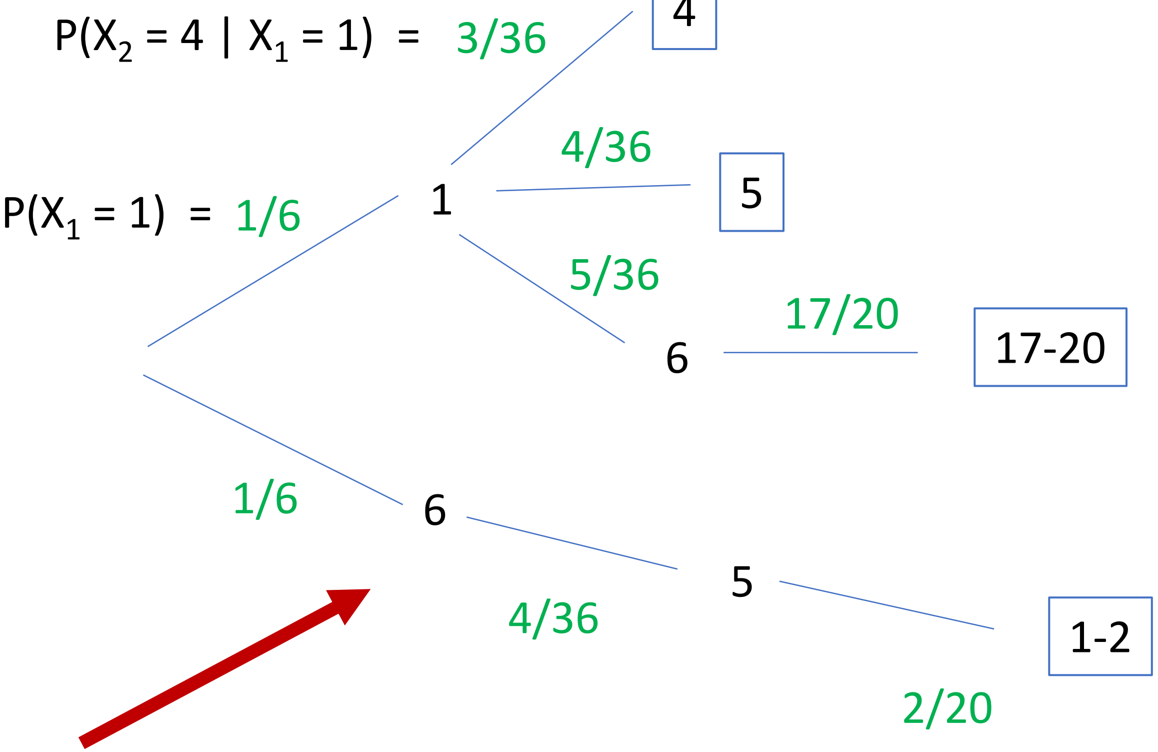
$P(X_2 = 4 \mid X_1 = 1) \cdot P(X_1 = 1) =$

0.0139

X_1

X_2

X_3



Can we fill in the probability notation for the lower part of the tree where $X_1 = 6$?

Using probability notation with tree diagrams

$P(X_1 = 1, X_2 = 4) =$

$P(X_1 = 1) \cdot P(X_2 = 4 \mid X_1 = 1) =$

0.0139

X_1

X_2

X_3

$P(X_1 = 1) = 1/6$

$P(X_2 = 4 \mid X_1 = 1) = 3/36$

1 → $\boxed{4}$ (3/36); 1 → $\boxed{5}$ (4/36): 0.0185; 1 → 6 (5/36) → $\boxed{17\text{-}20}$ (17/20): 0.0197

$P(X_1 = 6) = 1/6$

6 → 5 → $\boxed{1\text{-}2}$

$P(X_2 = 6 \mid X_1 = 6) = 4/36$

$P\,(X_3 = \{1,2\} \mid X_2 = 6, X_1 = 6) = 2/20$

$P\,(X_1 = 6, X_2 = 6, X_3 = \{1,2\}) =$

$P(X_1 = 6) \cdot P(X_2 = 6 \mid X_1 = 6\,) \cdot P(X_3 = \{1,2\} \mid X_1 = 6, X_2 = 6) =$ 0.00185